Development Methodology: Scrum

Sprint 1 Deliverable

1. Sprint Backlog – Sprint 1

(a) Module Selected for Sprint 1

Module Name:

4.1 Account Management Module

Module Description:

The Account Management module is responsible for handling user registration, account approval and deletion, and role-based authentication. This module serves as the foundation of the Library Management System because all other system functionalities depend on authenticated and authorized users.

Features Included in Sprint 1:

- Student registration (pending admin approval)
- Account deletion request
- Admin approval/rejection of accounts
- Role-based login system

(b) User Stories Selected for Sprint 1

US-01 – Student Registration Request

Story ID: US-01
Story Title: Student Registration
Importance: High
Estimate: 3 hours

User Story:
As a Student,
I want to request account creation,
So that I can access the library management system.

Sub User Stories:

- US-01.1: Design student registration form
- US-01.2: Validate input fields (email, password, student ID)
- US-01.3: Store registration request as “Pending”
- US-01.4: Display confirmation message after submission
- US-01.5: Prevent duplicate email registration

US-02 – Admin Approval/Rejection of Account

Story ID: US-02
Story Title: Admin Account Approval
Importance: High
Estimate: 3 hours

User Story:
As an Admin,
I want to approve or reject account creation requests,
So that only valid users gain access to the system.

Sub User Stories:

- US-02.1: Display list of pending account requests
- US-02.2: Provide Approve button
- US-02.3: Provide Reject button
- US-02.4: Update account status in database
- US-02.5: Notify student of approval/rejection

US-03 – Account Deletion Request

Story ID: US-03
Story Title: Account Deletion Request
Importance: Medium
Estimate: 2 hours

User Story:
As a User,
I want to request account deletion,
So that my account can be removed securely.

Sub User Stories:

- US-03.1: Provide “Request Account Deletion” option
- US-03.2: Store deletion request as pending
- US-03.3: Notify admin of deletion request
- US-03.4: Show confirmation message to user

US-04 – Role-Based Login System

Story ID: US-04
Story Title: Role-Based Login
Importance: High
Estimate: 4 hours

User Story:
As a User,
I want to log into the system based on my role,
So that I can access features specific to my permissions.

Sub User Stories:

- US-04.1: Create login page
- US-04.2: Validate credentials
- US-04.3: Restrict login if account status is “Pending”
- US-04.4: Redirect user to role-specific dashboard
- US-04.5: Display error message for invalid login

(c) Structured Specifications (Four User Stories)

Structured Specification – US-01

Student Registration

Primary Actor: Student
Secondary Actor: Admin

Preconditions:

- Student is not already registered.
- System is operational.

Main Flow:

1. Student opens registration page.
2. Student enters required details.
3. System validates inputs.
4. System checks for duplicate email.
5. System stores account as “Pending Approval.”
6. Confirmation message is displayed.

Alternate Flow:

- If email already exists → System displays error message.
- If fields are empty → System prompts user to complete required fields.

Postconditions:

- Registration request stored in database.
- Admin can view the request in pending list.

Structured Specification – US-02

Admin Account Approval/Rejection

Primary Actor: Admin
Secondary Actor: Student

Preconditions:

- Admin is logged in.
- There are pending registration requests.

Main Flow:

1. Admin opens pending requests dashboard.
2. Admin selects a student request.
3. Admin clicks Approve or Reject.
4. System updates account status.
5. Student is notified of decision.

Alternate Flow:

- If system error occurs → Display error message.
- If request already processed → Prevent duplicate action.

Postconditions:

- Account becomes Active (if approved) or Removed (if rejected).

Structured Specification – US-03

Account Deletion Request

Primary Actor: User
Secondary Actor: Admin

Preconditions:

- User is logged in.

Main Flow:

1. User selects “Request Account Deletion.”
2. System confirms action.
3. Request is stored as pending.
4. Admin receives notification.

Alternate Flow:

- If user cancels request → No action taken.

Postconditions:

- Deletion request appears in admin dashboard.

Structured Specification – US-04

Role-Based Login System

Primary Actor: User

Preconditions:

- User account exists.
- Account status is Active.

Main Flow:

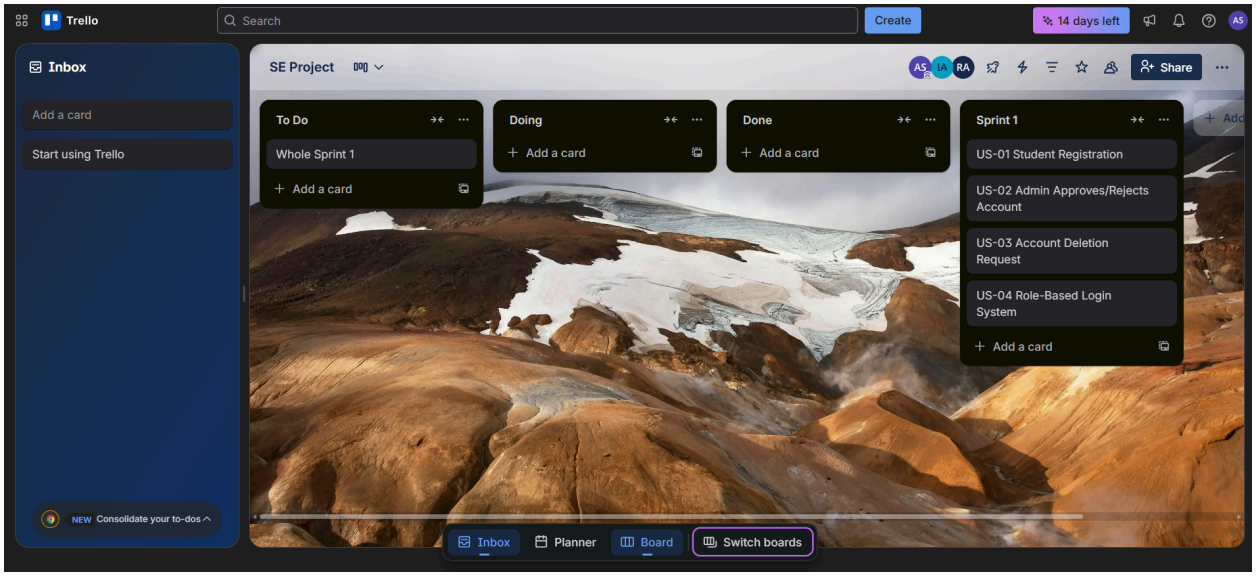
1. User enters email and password.
2. System validates credentials.
3. System checks account status.
4. System identifies user role.
5. User is redirected to role-specific dashboard.

Alternate Flow:

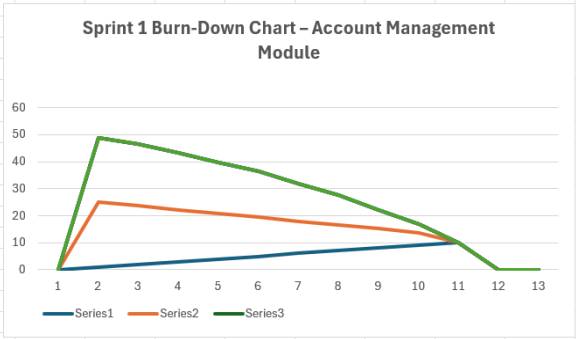
- Invalid credentials → Display error message.
- Account pending approval → Display “Awaiting Approval” message.

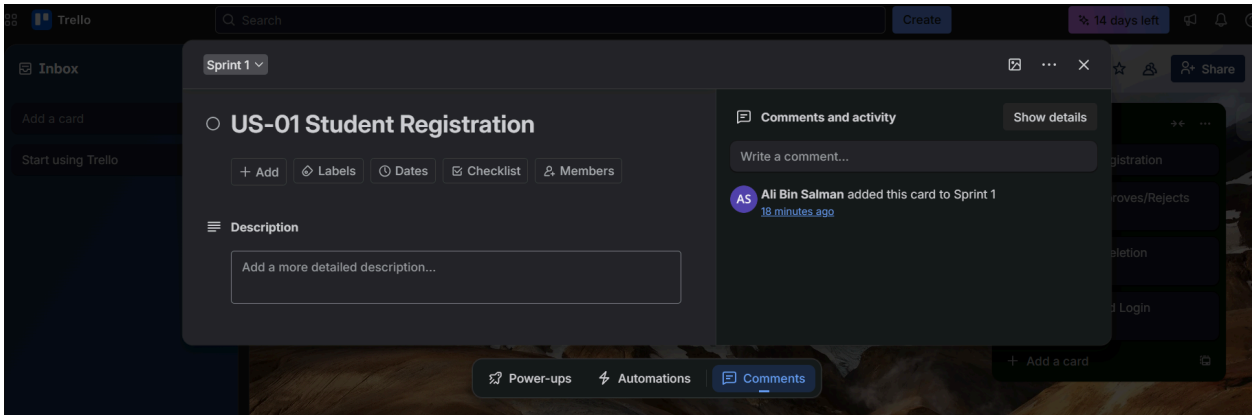
Postconditions:

- User successfully logged into role-based dashboard.



Day	Planned Tasks Remaining	Actual Tasks Remaining
1	24	24
2	21.6	23
3	19.2	21
4	16.8	19
5	14.4	17
6	12	14
7	9.6	11
8	7.2	7
9	4.8	3
10	0	0





Sprint 1 Summary

Sprint 1 focuses on establishing a secure and controlled account management system. This module ensures that:

- Only approved users gain access.
- Admin maintains system control.
- Authentication is secure and role-based.
- The system foundation is prepared for future modules